

List of Exercises: FDS/IDS

October 27th, 2025

1. In a given application, we have information about the ages of a set of 12 people. Their values are 12, 30, 24, 10, 10, 23, 43, 67, 79, 34, 56, 51.
 - a) What is the median of these ages? Explain.
 - b) What is the mode of these ages? Explain.
 - c) How would you obtain the difference between the 99% percentile and the 10% of this set of ages, in Python? How about the 1% percentile?
 - d) What are the results of normalizing and standardizing these data?
2. Suppose that we add two more age values to the set mentioned in item (1): 10 months and 100 years.
 - a) Apply normalization and standardization to this new set of data.
 - b) Should you give any preference to apply normalization or standardization to this new set of data?
3. Answer the following questions:
 - a) What is the objective of boxplot graphs?
 - b) What are the functions of the spread measures: “range” and “interquartile range”. Is there any advantage of using one over the other?
 - c) What other spread measures can we use to analyse data?
4. Figure 1 shows a “scatterplot”. What information can you infer from this graph?
5. Figure 2 shows a “parallel plot”. What kind of information can you infer from this graph?
6. When visualizing data, it may be important to reorder/rearrange variables or sort variable values. Give an example, where this order can yield a better visualization than the one over the original, not rearranged data.

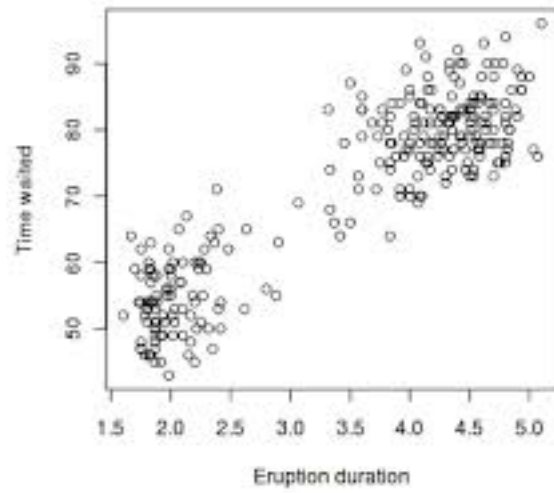


Figure 1: Scatterplot Example

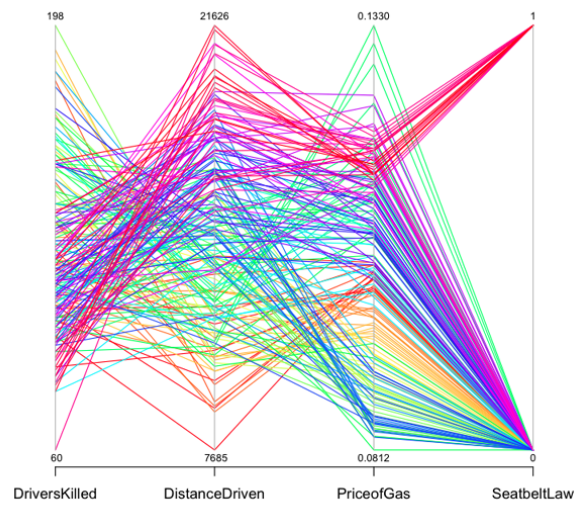


Figure 2: Parallel plot Example

7. The “nearest neighbours” strategy can also be used to “impute” values to unknown variable values. Explain how you can use nearest neighbours to impute missing variable values.
8. What is the main idea behind PCA – Principal Component Analysis and why is it useful?
9. Suppose you are given a CSV (Comma-Separated Values) table. When you read this table using the Python function `read_csv`, what are the variable types stored internally for a variable that has only two values? How about when you read the same data in the WEKA software or R?
10. What are the basic variable types used in data analysis?
11. Explain the difference between the distance calculated using “simple matching” and the “Jaccard” distance. In what situation, we apply one or the other?
12. What is the difference between Pearson correlation and simple linear regression?
13. Consider the following data table:

Inst/Var	V1	V2
I1	1.5	1.7
I2	2	1.9
I3	1.6	1.8
I4	1.2	1.5

Given a new observation (1.4,1.6), which two observations in the table are nearer the new data point, using the Euclidean distance?

14. Which approach would you use to plot a histogram for a continuous numeric variable?
15. What is the objective of data sampling?